

ARTIFICIAL INTELLIGENCE

The Third Revolution

What AI is, what it is not, and what it means for those who teach

A session for the educators of Pranavananda International School, Gurugram

Basab Dasgupta

AI will never...

Keep a private count: how many of these have already happened?

1



Beat the world chess champion

2



Beat the world champion at Go

3



Solve a 50-year-old problem in biology

4



Win gold at the International Maths Olympiad

5



Drive a taxi with nobody at the wheel

6



Write an essay you cannot tell from a student's

7



Know that it exists

8



Decide what is right for a child

Six of the eight have already happened

How many did you count?

1



Beat the world chess champion

✓ 1997 · Deep Blue

2



Beat the world champion at Go

✓ 2016 · AlphaGo

3



Solve a 50-year-old problem in biology

✓ 2020 · AlphaFold

4



Win gold at the International Maths Olympiad

✓ 2025 · gold medal

5



Drive a taxi with nobody at the wheel

✓ Since 2020 · Waymo

6



Write an essay you cannot tell from a student's

✓ Since 2022

7



Know that it exists

? Nobody knows

8



Decide what is right for a child

✗ No

Every line we drew has moved.

So which lines will hold?
And what should that change about how we teach?

Today, in five parts

90 minutes. Questions are welcome at any point.

- | | | |
|----|---|--------|
| 01 | What is "AI" anyway?
Rules or examples, and the ladder from data to wisdom | 18 min |
| 02 | How the machine learned to talk
Seventy years in one line, and what really happened in 2022 | 13 min |
| 03 | The reality check, from the staffroom
Two live demonstrations: invented sources and hidden bias | 20 min |
| 04 | The human edge in teaching
What stays human, and assessment when every student has AI | 14 min |
| 05 | Explaining AI, and what comes next
The new CBSE curriculum, AI by stage, and who we are | 12 min |

PART 01

What is "AI" anyway?

Why the word confuses everyone, and a better way to think about it.

0:08 – 0:26



Intelligence is what you use when you don't know what to do.

Widely attributed to Jean Piaget

The question for today

Is AI intelligent by this definition? When it meets something it was never trained on, does it know what to do?

One word, many meanings

The confusion behind the buzzword



A car that drives itself



A chatbot that writes circulars



An app that picks your next video

All three are called "AI". They have very little in common. Marketing adds the rest: hype for some, fear for others.

A working definition

A system that can perceive its environment, process information, learn from patterns, and make decisions to reach a goal, often in situations it was never explicitly programmed for.

AI is a new technology, not "better IT"

Three revolutions, three different kinds of capability

The common explanation: "AI exists because we now have more data and faster computers." About 10% true, and 90% misleading.

Electricity

From the 1880s

Energy becomes motion,
light and heat

≠

Information technology

From the 1950s

Information is stored,
processed and sent

≠

Artificial intelligence

From the 2010s

Data becomes prediction,
understanding and creation

Nobody calls the computer "better electricity". AI is not "better IT".

Rules or examples?

The single most useful idea for explaining AI



Traditional software

A person writes the rules. The computer follows them.

Example: adding up a report-card total.



Machine learning

A person gives thousands of examples. The computer finds the rules itself.

Example: reading a child's handwriting.

Machine learning is closer to how a child learns than to how a textbook is written.

The ladder from data to wisdom

Using a Class 7 maths unit test

Wisdom

Reteach fractions with paper folding. Call one parent, because you know something is wrong at home.

Knowledge

The three students below 40 were all absent for the fractions lesson.

Information These are Class 7 maths unit-test marks.

Data

42, 67, 35, 88, 31, 74

Could an AI
have known to
call that one
parent?

Where does AI sit? Where do we?

The ladder, read honestly in 2026



Strongest between information and knowledge. It processes vast data and finds patterns we would never spot.



At the top. Deciding what matters, what is fair, what is kind, and what is right for this child.

AI now suggests actions too. It does not know which one matters, or why.

PART 02

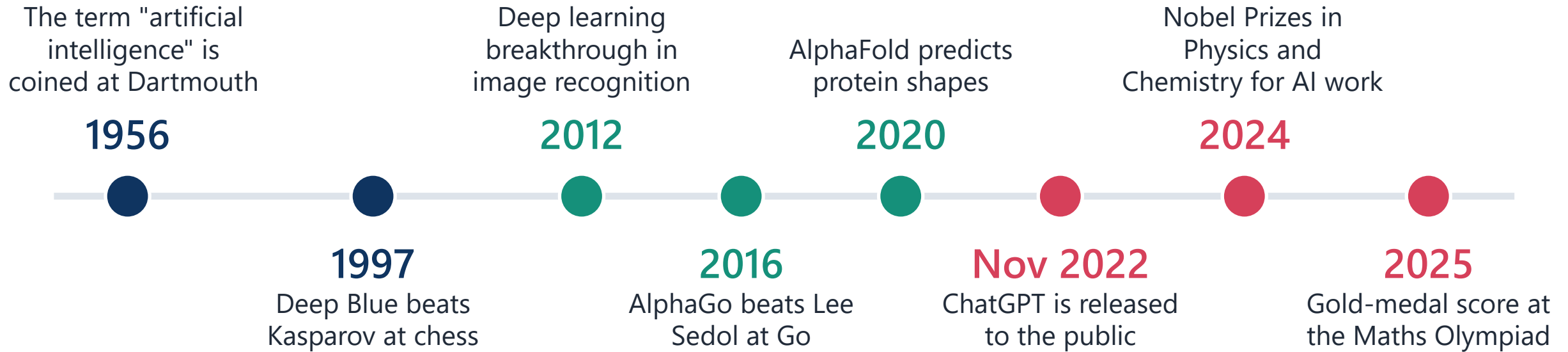
How the machine learned to talk

From chess to conversation, and what really happened in 2022.

0:26 – 0:39

Seventy years in one line

Slow for forty years. Then every two or three years. Then every year.

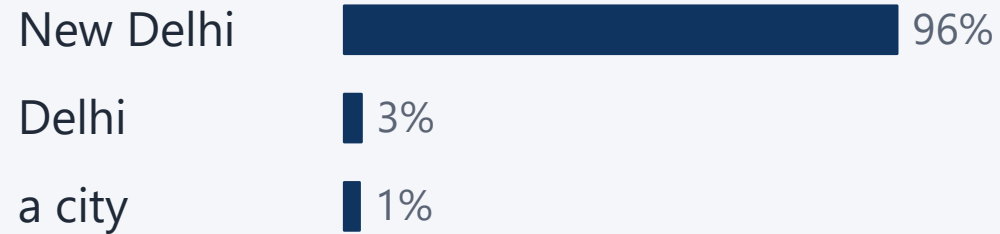


Forty-one years between the first two dots. One year between the last two.

At heart, it guesses the next word

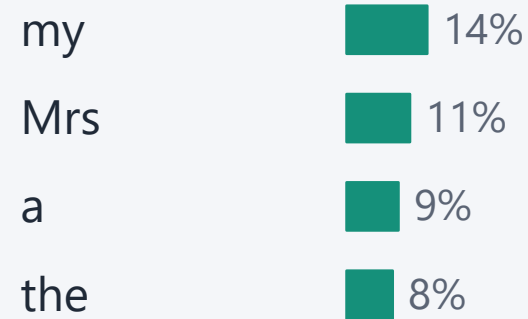
The same engine produces facts and fiction

The capital of India is ____



One continuation dominates. So it usually gets the fact right.

The best teacher I ever had was ____



Thousands of good continuations. So it can write a moving essay.

Illustrative percentages

It picks what is likely, not what is true. When the likely answer is false, it says it just as confidently.

The breakthrough: language

Large language models cracked linguistic intelligence, not all of intelligence

Why language was crackable

Language is full of patterns: grammar, word order, which words go together. With enough text, a machine can learn them statistically.

What was achieved

Writing, translating, summarising and explaining, with astonishing fluency, in many languages.

The mistake

Treating fluent language as general intelligence. We make the same mistake with people: the most articulate student is not always the one who understands best.

Does it understand?

John Searle's "Chinese Room", 1980



A man who speaks no Chinese sits in a locked room with a huge rulebook. Chinese questions come under the door. He follows the rules and slides back perfect Chinese answers. Outside, everyone is convinced a Chinese speaker is inside.

He understands nothing.

One view

It is statistics, however impressive. There is nobody in the room.

Another view

The rulebook has become so rich that something like understanding has formed inside it.

Honest answer: genuinely contested.
Serious researchers sit on both sides.

If language was the first domino, what are the others?

Four kinds of intelligence, and where AI stands on each

Linguistic

Achieved

Reading, writing, translating, conversing

Reasoning

Moving fast

Planning and solving new problems.
"Thinking" models since 2024

Common sense

Patchy

Passes hard exams, then fails a question a seven-year-old would get right

Embodied

Early days

Moving and handling objects in the physical world

The world's bet: because language fell, the rest will follow. A bet, not a certainty.

PART 03

The reality check, seen from the staffroom

What goes wrong, and two things you can see happen in front of you.

0:39 – 0:59

Hallucination: confidently, fluently wrong

It predicts what is likely, not what is true

It can invent a source that looks completely real: authors, a year, a journal, even page numbers. Everything looks right. It simply does not exist.



New York, 2023

A lawyer asked ChatGPT for earlier court cases. It gave six. None existed. He was fined.



Bengaluru, December 2024

A tax tribunal order worth about ₹669 crore cited four judgments. None existed. It was recalled within a week.

Why it guesses: training tends to reward a confident answer over "I don't know".

Live: can we catch it inventing?

A demonstration, not an accusation. Web search switched off.

THE PROMPT

Without searching the web, list three published research papers on the effect of homework on primary school children in Haryana, with authors, year and journal.

- 1 Read the three references aloud
- 2 Open scholar.google.com
- 3 Paste one exact title, in quotes
- 4 Does it exist?

If it cannot be found

A reference list that looks perfect proves nothing. Check one.

If it refuses or finds real ones

Progress. The tools improve because people check.

Bias: it learns our habits, including the unfair ones

AI learns from what people wrote

A 2016 study of Google News text

Man is to computer programmer as
woman is to...

homemaker

Bolukbasi and others, NeurIPS 2016



Amazon's hiring tool

Trained on ten years of CVs, mostly from men. It learned to mark down CVs containing the word "women's". Amazon dropped it.

Reuters, October 2018

Nobody programmed either result. It was already in the data.

Live: the translator must choose "he" or "she"

Hindi does not say. English must.

वह डॉक्टर है।  ___ is a doctor.

वह नर्स है।  ___ is a nurse.

**Which word will
the machine
choose for each?**

Every AI-made story or worksheet makes this choice for you, many times over,
without telling you.

Why AI detectors cannot be trusted

Before any school buys one



OpenAI withdrew its own detector

Released January 2023. Withdrawn on 20 July 2023, citing its "low rate of accuracy".

61%

of essays by non-native English writers wrongly flagged as AI-written by seven popular detectors

Stanford study, Liang and others, 2023

Detectors flag simple, predictable writing. That is exactly how careful second-language writers write.

In India, almost every student writes English as a second language.

Deepfakes and student data

Two risks every school needs a plan for



Deepfakes

Anyone's face or voice can now be put on a video. A teacher's face is easy to find online.

Report: cybercrime.gov.in or helpline 1930.

Decide the school's response before it happens, not after.



Student data

The DPDP Act, 2023 requires verifiable parental consent before a child's personal data is processed.

Never paste a student's name, marks, health details or photo into a public AI tool. Write "a Class 6 student" instead.

PART 04

The human edge in teaching

Where we still lead, and what that means for teaching and assessment.

0:59 – 1:13

Where we still lead

Three things AI does not have



Purpose and values

AI can raise a class average efficiently. It cannot tell you whether that is the right goal this term.



Care

AI can detect sadness in a sentence. It does not feel. A child knows the difference between being noticed and being detected.



Lived experience

AI has read about rain. It has never walked to school in it.

Teachers decide the why: goals, values, what matters.

AI helps with the how: drafting, sorting, translating.

AI automates tasks, not jobs

A teacher's job is a very large bundle of tasks

AI can help with

A first draft of a question paper

One passage at three reading levels

A circular translated for parents

First-draft report-card remarks, which you correct

Stays human

Knowing each child

Judging effort and honesty

Noticing when something is wrong at home

Mentoring, and modelling values

Keeping children safe

Which column is the reason you became a teacher?

Assessment when every student has AI

Look at the process, not only the product



Drafts

Ask for the outline, the first draft and the final version.



Explain it

Two minutes: "Why did you choose this example?"



In-class work

Some writing happens in the room, by hand.



Reflection

"What did the AI get wrong, and how did you fix it?"

Grade the thinking, not just the answer.

Three questions before anyone shares AI-assisted work

The test every learner in our lab takes. It works for teachers too.



Can you understand it?

Could you explain every part, or fix it if it broke?



Can you explain it?

Were the ideas and choices yours?



Can you disclose it?

Can you say plainly what the AI did and what you did?

Disclosure is the line between using AI and passing off its work as your own.

PART 05

Explaining AI, and what comes next

It is now part of the curriculum. And some lines may never move.

1:13 – 1:25

It is now in the curriculum

CBSE Computational Thinking and AI framework, Classes 3 to 8, from 2026–27

Classes 3 to 5

Computational thinking woven into maths, languages and EVS through puzzles, games and stories.

Classes 6 to 8

Foundational AI concepts introduced alongside computational thinking.

For teachers

CBSE's training theme for the session:
"Computational Thinking and Understanding AI".

CBSE circular Acad-15/2026, 1 April 2026

Explaining AI is no longer only the computer teacher's job.

How to explain AI, stage by stage

One idea per stage, and one rule for all of them

Classes 1 to 5

"The machine learns from examples, just like you."

Sorting game: the children are the machine. Then slip a red apple into the green pile.

Classes 6 to 8

"It guesses the next word."

The phone keyboard suggests the next word. If it is guessing, can it be wrong?

Classes 9 to 12

"It predicts. It does not look up."

Then the harder talks: bias, deepfakes, and three questions before sharing.

For every stage: let children catch the machine making a mistake.

Which lines might hold?

The frontiers AI has not crossed



General intelligence

One system that does everything a person can, with the same flexibility. Not yet. Nobody knows when.



Self-awareness

Does it know it exists? We cannot fully answer that about ourselves.



Sustainability

Data centres used about 415 TWh in 2024; expanding to 945 TWh by 2030: more than all of Japan uses today (IEA).



Judgement and ethics

AI can tell you what is likely. It cannot tell you what is right.

The last line is the one teachers stand on every day.

Learn it. Guide it. Use it well.

Three things to take back to the staffroom

1

Learn it

Twenty minutes a week on your own subject. Try to catch it out.

2

Guide it

Decide what AI may do in your classroom, and tell students clearly.

3

Use it well

AI amplifies what it is given: good teaching, or laziness and bias. Stay the one asking questions.

Go further: futureskillsprime.in, MeitY's platform for courses in AI

Bharat Tech-Shakti Mission

Swami Pranavananda AI & Robotics Lab, Bharat Sevashram Sangha, New Delhi



Who we are

A group of professionals and volunteers who run a hands-on AI and robotics lab at Bharat Sevashram Sangha.



What we do

Anyone can walk in and learn AI by doing it. A nominal fee, no entrance test, no age limit.



Why we do it

Most people do not use AI tools. Not because they are expensive. Because nobody has shown them how.

सृजन से शक्ति

Carrying forward the founder's belief that service means helping people stand on their own feet.

Learn with us, and join us

Free, bite-sized, and open to everyone



AI Shakti bite-sized learning

Free. One short lesson at a time, on WhatsApp.



Kindly Verify, our blog

Free. Real stories of AI in Indian lives: bharattechshakti.org/blog



The AI Shakti book

Our beginner course: first question to first circuit.



Coming soon: Gurugram

AI community learning at Bharat Sevashram Sangha.

Join us

Teach a session, help translate lessons, or bring your students to the lab.

BharatTechShakti@gmail.com · +91 96256 77741 · bharattechshakti.org

AI is the tool.
Wisdom is the guide.
Humanity is the purpose.

Questions?

Basab Dasgupta · dasgupta.basab@gmail.com · +91 98119 07741